Gloria Pietropolli

Date of Birth: 15 February 1996
Nationality: Italian

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Research Interests

- Design and analysis of machine learning algorithms, with a focus on interpretable model discovery.
- Application of interpretable machine learning across diverse domains, including reinforcement learning with large language models, cryptography, control systems, and data-driven economic modeling.
- Development of deep learning models for environmental and oceanographic data.

Education and Work Experience

Apr 2024–present **Postdoctoral Researcher**, University of Trieste, Italy
Under the supervision of Prof. Luca Manzoni.
Scientific-disciplinary sector (SSD): INF/01
Expected duration 24 months (from 05/04/2024 to 04/04/2026)

Jan–Mar 2024 **Independent Research Contract**, University of Trieste, Italy
Experimental evaluation of optimization algorithms based on the computational paradigm of cellular automata. Under the supervision of Prof. Eric Medvet.

Nov–Dec 2023 **Independent Research Contract**, University of Trieste, Italy
Project on integrated pathways for diagnosis, therapy, and assisted care of multimorbid individuals at risk of atrial fibrillation. Under the supervision of Prof. Luca Bortolussi.

Nov 2020–Oct 2023 **Ph.D. in Computer Science**, University of Trieste, Italy
Ph.D. Program: Earth Science, Fluid-Dynamics, and Mathematics. Interactions and Methods; Title: *Machine Learning Applications to Data Reconstruction in Marine Biogeochemistry*. Under direction of Prof. Luca Manzoni & Dr. Gianpiero Cossarini.
The Ph.D. was obtained on March 21, 2024. Disciplinary area: INF/01.
Mark: Excellent cum laude.

Oct 2018-Jul 2020 **M.Sc. in Mathematics**, University of Padua, Italy
Under the supervisor of Prof. Antonia Larese.

Oct 2015-Jul 2018 **B.Sc. in Mathematics**, University of Padua, Italy
Under the supervisor of Prof. Marco Favretti.

Research Abroad

Sep 2025 **NOVA University, Lisbon, Portugal**, Duration: 1 month.
Explainable Deep Learning for Oceanographic Applications. Scientific coordinator: Prof. Mauro Castelli.

Sep 2024 **NEOMA Business School, Paris, France**, Duration: 1 week.
Optimization algorithm for automatic detection of structural relations in structural equation modeling. Scientific coordinator: Prof. Laura Trinchera.

Apr–Jun 2024 **NOVA University, Lisbon, Portugal**, Duration: 3 months.
Investigating interpretable genetic programming optimization techniques. Scientific coordinators: Prof. Mauro Castelli and Prof. Leonardo Vanneschi.

Jan 2024 **MIT, Cambridge, MA, USA**, Duration: 1 month.
Integrating evolutionary optimization techniques with large language models. Scientific coordinator: Prof. Una-May O'Reilly.

Sep 2022 **Université Nice Sophia Antipolis, Nice, France**, Duration: 1 week.
Scientific coordinator: Prof. Enrico Formenti.

Summary of Publications

The rank and quartile reported follow the classifications provided by Scopus and CORE.

- 12 Publications in peer-reviewed international journals. 8 Q1; 4 Q2
- 2 Book chapters
- 11 Publications in peer-reviewed international conferences. 5 A-ranked; 5 B-ranked
- 2 Publications in peer-reviewed international workshops
- 9 Abstract and local proceedings publications

Research Profile

Research Summary

My first line of work focuses on interpretable machine learning and automated model discovery. Examples include the development of methodologies for generating interpretable agents in reinforcement learning tasks (e.g., Atari games) using large language models [C6, under review], models for automatically generating cryptographic Boolean functions with compact and interpretable representations [C10, work in progress], and models for designing interpretable control laws aligned with optimal control references [J8]. I am currently developing automated search methods in composite-based Structural Equation Modeling (SEM), in collaboration with NEOMA, Twente, and NOVA universities, for data-driven economic modeling [under review]. I also study machine learning algorithms from a theoretical perspective, implementing methodological improvements to improve their performance in practical applications [J3, J2, C11, C9, C8, C7, C4].
My second line of work applies machine learning to environmental and oceanographic sciences through an active collaboration with the National Institute of Oceanography and Applied Geophysics (OGS). I develop deep learning models for oceanographic data, including 3D modeling and uncertainty-aware predictions [C1]. I developed a model for biogeochemical marine variable prediction [J5], which has been applied to data assimilation in the Mediterranean Sea [J7]. I have also developed a model for biogeochemical profile reconstruction [J9]. This work laid the foundation for the GLOBIO project, where I serve as WP Leader, and produced models now embedded in the Copernicus Marine Service for operational use.

Pages and bibliometrics

Google Scholar `https://scholar.google.com/citations?user=-iUZFA4AAAAJ&hl=it`
Citations: 164, h-index: 9

Scopus `https://www.scopus.com/authid/detail.uri?authorId=57642608800`
Citations: 86, h-index: 6

Orcid `https://orcid.org/0000-0001-7623-8419`

Dblp https://dblp.org/pid/318/8329.html

Research Community Services

Chairing & Organizing Roles

GECCO 2025 **Women+ @ GECCO Workshop Chair**, ACM Genetic and Evolutionary Computation Conference. 14-18 July; Malaga, Spain.

Ital-IA 2025 **Workshop Chair**, Convegno Nazionale CINI sull'Intelligenza Artificiale - AI and Sustainability WS. 23-24 June; Trieste, Italy.

EuroGP 2025 **Local Chair**, The Leading European Event on Bio-Inspired AI. 23-25 April; Trieste, Italy.

EvoStar 2023 **Student Affairs and Student Workshop Chair**, The Leading European Event on Bio-Inspired AI. 12-14 April; Brno, Czech Republic.

Automata 2023 **Organizing Committee Chair**, The 29th International Workshop on Cellular Automata and Discrete Complex Systems. 30 Aug-1 Sep; Trieste, Italy.

CMC 2022 **Organizing Committee Chair**, 23rd Conference on Membrane Computing. 6-9 Sep; Trieste, Italy.

Program Committe Member

2024, 2025 **ECAI**, European Conference on Artificial Intelligence.

2025 **GECCO**, ACM Genetic and Evolutionary Computation Conference, Genetic Programming track.

2025, 2026 **EuroGP**, The Leading European Event on Bio-Inspired AI - EuroGP track.

2024 **ALife**, The Conference on Artificial Life

Refereeing Duties

Journals Genetic Programming and Evolvable Machines, AI Communications, IEEE Transactions on Artificial Intelligence, Journal of Cellular Automata, Natural Computing, Data Mining and Knowledge Discovery.

Conferences Combinatorics on Words (WORDS 2023), European Symposium on Algorithms (ESA 2023), European Conference On Artificial Intelligence (ECAI 2024, 2025), The Conference on Artificial Life (ALIFE 2024, 2025), ACM Genetic and Evolutionary Computation Conference (GECCO 2025), The Leading European Event on Bio-Inspired AI (EuroGP 2025, 2026).

Other

Jun 2025- present Science Communication Vice-Coordinator for ROAR-NET (Randomised Optimisation Algorithms Research Network).

May 2025- present Working Group Member for ROAR-NET (Randomised Optimisation Algorithms Research Network

Teaching Activities

Courses in which I was the <u>course leader</u> are <u>underlined</u>.

Invited Lecturer

A.Y. 2025/2026 **Deep Learning**, NOVA University, PhD in Information Management (2 classes, 4 hrs.) Invited by Professor Mauro Castelli, the course leader.

Lecturer

A.Y. 2025/2026 <u>**Introduction to Computer Programming**</u>, University of Trieste, BSc in Artificial Intelligence and Data Analytics (20 hrs.)

A.Y. 2024/2025 **Population-Based Optimization Methods (Ph.D. course)**, University of Trieste, Ph.D. in Applied Data Science & Artificial Intelligence (20 hrs.)

A.Y. 2024/2025 **Introduction to Computer Programming**, University of Trieste, BSc in Artificial Intelligence and Data Analytics (20 hrs.)

A.Y. 2023/2024 **Population-Based Optimization Methods (Ph.D. course)**, University of Trieste, Ph.D. in Applied Data Science & Artificial Intelligence (16 hrs.)

A.Y. 2023/2024 **Introduction to Computer Programming**, University of Trieste, BSc Artificial Intelligence and Data Analytics (20 hrs.)

Teaching Assistant

A.Y. 2025/2026 **Algorithms and Data Structures**, University of Trieste – BSc in Artificial Intelligence and Data Analytics (20 hrs.)

A.Y. 2024/2025 **Algorithms and Data Structures**, University of Trieste – BSc in Artificial Intelligence and Data Analytics (20 hrs.)

Tutor

A.Y. 2022/2023 **Algorithms and Data Structures**, University of Trieste, BSc in Artificial Intelligence and Data Analytics (50 hrs.)

A.Y. 2021/2022 **Algorithms and Data Structures**, University of Trieste, BSc in Artificial Intelligence and Data Analytics (50 hrs.)

A.Y. 2021/2022 **Computer Architecture**, University of Trieste, BSc in Artificial Intelligence and Data Analytics (50 hrs.)

A.Y. 2020/2021 **Computer Programming**, University of Trieste, BSc in Mathematics (50 hrs.)

A.Y. 2019/2020 **Linear Algebra and Geometry**, University of Padua, BSc in Environmental and Spatial Engineering (50 hrs.)

A.Y. 2018/2019 **Numerical Analysis**, University of Padua, BSc in Energy Engineering (50 hrs.)

A.Y. 2018/2019 **Calculus I**, University of Padua, BSc in Physics & Astronomy (50 hrs.)

Student Supervision

- **Ph.D.**
  - Co-supervisor with Prof. Luca Manzoni and Dr. Gianpiero Cossarini of Ph.D. student Teresa Tonelli, since Nov 2023. Research topic: "Machine learning in marine science: model and data fusion for prediction, interpolation, and extrapolation." Collaboration between the University of Trieste and the National Institute of Oceanography and Applied Geophysics (OGS).
- **MSc**
  - Co-supervisor with Prof. Luca Manzoni of Marco Carollo, MSc in Data Science and Scientific Computing, University of Trieste, Mar 2025. Thesis: "Interpolation of biogeochemical variables on Mediterranean data using evolutionary techniques."
  - Co-supervisor with Prof. Luca Bortolussi and Dr. Giovanni Baj of Filippo Olivo, MSc in Data Science and Scientific Computing, University of Trieste, Mar 2024. Thesis: "Design and optimization of a tiny deep learning model for atrial fibrillation detection."
  - Co-supervisor with Prof. Luca Manzoni and Prof. Mauro Castelli of Ines Ferreria, MSc in Information Management, NOVA University, Portugal, Jan 2024. Thesis: "Dynamic modeling of 3D marine fields: unleashing the power of convolutional transformers for advanced environmental analysis."
  - Co-supervisor with Prof. Luca Manzoni and Dr. Gianpiero Cossarini of Teresa Tonelli, MSc in Mathematics, Sep 2023. Thesis: "Genetic programming to reconstruct alkalinity distribution in the Mediterranean Sea."
- **BSc**
  - Co-supervisor with Prof. Luca Manzoni of Jury Paglierani, BSc in Physics, University of Trieste, Dec

2021. Thesis: "Approcci alla simulazione di sistemi fisici mediante l'utilizzo di reti neurali."

Funded Projects, Awards, and Scholarships

Research Projects

Aug 2025 — Winner of a three-year postdoctoral fellowship at the Centrum Wiskunde & Informatica in Amsterdam to lead the WP17 "Explainable decision models and user interface" within the project "New strategies for the early diagnosis, risk stratification and co-management of familial hypercholesterolemia", directed by Peter Bosman and Tanja Alderliesten. Expected start date: January 1, 2026.

Sep 2024-present — **WP Leader**, Member and Workpackage Leader for WP2 ("Global model contruction") of the research project "Bridging Global and Local Scales for Biogeochemical Profiles Prediction". Copernicus Marine Service Evolution.

Apr 2024-present — **WP Leader**, Member and Workpackage Leader for WP3 ("Engineering of a tool to generate CA for cryptographic primitives") of the research project "Cellular Automata Synthesis for Cryptography Applications (CASCA)". PRIN 2022 PNRR.

Grants

Sep 2025 — **FCT Mobility Grant**, Project: Explainable Deep Learning for Oceanographic Applications. The grant supported a month visit to the NOVA University in Lisbon (Sep 2025, amount financed: **€1965**).

Sep 2024 — **ROAR-NET Short-Term Scientific Mission Grant**, Project: Optimization Algorithm for Automatic Detection of Structural Relations in Structural Equation Modeling. The grant supported one-week visit to the NEOMA University in Paris (Sep 2024, amount financed: **€1430**).

Oct 2024 — **SPECIES Scholarship**, The scholarships supported a three-month in-person internship at a chosen research institution (amount financed: **€2700**).

June 2023 — **PI of the ISCRA class C project IsCa8-MEDConTr**, Project IsCa8-MEDConTr; **80000 core processor hours** awarded by CINECA on the supercomputer LEONARDO.

July 2023 — **ACM-W Scholarship**, To participate in the Genetic and Evolutionary Computation Conference (GECCO 2023) July 15 - 19, 2023, Lisbon, PT.

Awards

July 2024 — Best Paper Nomination at the GECCO Conference for the paper "Large Language Model-based Test Case Generation for GP Agents".

April 2023 — Best Paper Nomination at the Eurogp Conference for the paper "A Self-Adaptive Approach to Exploit Topological Properties of Different GAs' Crossover Operators".

Science Outreach and Education

2022–2024 — **High School Outreach Courses – Scientific Education**, Trieste, Italy, 20-hour project-based outreach courses on advanced AI topics for high school students. In 2022, the focus was on clustering and recommending systems. In 2023 and 2024, the focus was on convolutional neural networks and imaging.

2020–2021 — **"Coding Girls" Project – High School Teacher of Computer Science**, Trieste, Italy, Project focused on fostering equal opportunities in the scientific and technological sector.

Schools

Nov 3-6, 2025 — **Invited Participant at the Workshop**, *Learned Methods for Operations Research*, Amsterdam, The Netherlands.

Nov 25-29, 2022 **PhD School Participant**, *6th International Gran Canaria School on Deep Learning*, Las Palmas de Gran Canaria, Spain.

Nov 19-23, 2021 **PhD School Participant**, *4th Advanced Online & Onsite Course on Data Science & Machine Learning*, Siena, Italy (attended online).

Talks Given

Conference and Workshop presentations:

- May 2025. Presentation of "GLOBIO: Bridging Global and Local Scales for Biogeochemical Profiles Prediction" at European Geoscience Union General Assembly, Wien, Austria.
- Apr 2025. Presentation of "Introducing Crossover in SLIM-GSGP" at EuroGP25, Trieste, Italy.
- Jul 2024. Presentation of "Large Language Model-based Test Case Generation for GP Agents" at GECCO24, Melbourne, Australia.
- Jul 2024. Presentation of "The Role of the Substrate in CA-based Evolutionary Algorithms" at GECCO24, Melbourne, Australia.
- May 2023. Presentation of "A convolutional deep learning approach for the reconstruction of nutrient and carbonate system variable profiles" at 54th International Liege Colloquium On Ocean Dynamics, Liege, Belgium.
- Apr 2023. Presentation of "A Self-Adaptive Approach to Exploit Topological Properties of Different GAs' Crossover Operators" at EuroGP23, Brno, Czech Republic.
- Dec 2022. Presentation of "GANs for integration of deterministic model and observations in marine ecosystems" at the 5th workshop HPC-TRES, Trieste, Italy.
- Sep 2022. Presentation of "P Systems inference via Grammatical Evolution" at CMC22, Trieste, Italy.
- Sep 2022. Presentation of "GANs for Integration of Deterministic Model and Observations in Marine Ecosystem" at EPIA22, Lisbon, Portugal.
- Apr 2022. Presentation of "Combining geometric semantic gp with gradient-descent optimization" at EuroGP22, Madrid, Spain.
- Dec 2021. Presentation of "Multivariate relationship in big data collection of Ocean Observing System" at the 4th workshop HPC-TRES, online.

Seminars

- April 2023. "Machine Learning methods for Oceanographic applications" National Institute of Oceanography and Applied Geophysics, Trieste, Italy.
- December 2021. "Combining Gradient Descent and GP for better symbolic regression" DSSC Seminar, University of Trieste, Trieste, Italy.
- March 2021. "Simulation of complex physical systems with Graph Neural Network" DSSC Seminar, University of Trieste, Trieste, Italy.

Journal Papers (peer reviewed)

The list of publications is provided together with the current rank of the journal in Scopus. An asterisk (*) indicates multiple first authors with equal contribution. If the order of authors is alphabetical, this is indicated by (a) at the beginning.

[J12] S. Jorgensen*, G. Nadizar*, G. Pietropolli*, L. Manzoni, E. Medvet, U. M. O'Reilly, E. Hemberg. "Policy Search through Genetic Programming and LLM-assisted Curriculum Learning." *ACM Transactions on Evolutionary Learning and Optimization*, 2025 DOI:10.1145/3772718
Scopus: Computer Science, quartile Q1

[J11] (a) D. Farinati, G.Pietropolli, L. Vanneschi. "A Study on the Dynamics and Effectiveness of the Deflate Geometric Semantic Mutation." *IEEE Transactions on Evolutionary Computation*, 2025 DOI:10.1145/3772718
Scopus: Theoretical Computer Science, quartile Q1

[J10] L. Rovito, L. Bonin, D. Farinati, L. Vanneschi, L. Manzoni, A. De Lorenzo, G. Pietropolli. "Semantic-based recombination and mutation in cellular-inspired genetic programming." *Genetic Programming and Evolvable Machines*, 2025 DOI:10.1007/s10710-025-09524-7
Scopus: Computer Science Applications, quartile Q2

[J9] G. Pietropolli, L. Manzoni, G. Cossarini. "PPCon 1.0: Predict Profiles Convolutional, a Novel Deep Learning Approach for Smooth Nutrient Profile Predictions." *Geoscientific Model Development*, 2024 DOI:10.5194/gmd-17-7347-2024
Scopus: Modeling and Simulation, quartile Q1

[J8] F. Marchetti, G. Pietropolli, F. J. Camerota Verdù, M. Castelli, E. Minisci. "Automatic Design of Interpretable Control Laws Through Parametrized Genetic Programming with Adjoint State Method Gradient Evaluation." *Applied Soft Computing*, 2024. DOI: 10.1016/j.asoc.2024.111654
Scopus: Software, quartile Q1

[J7] C. Amadio, A. Teruzzi, G. Pietropolli, L. Manzoni, G. Coidessa, G. Cossarini. "Combining Neural Networks and Data Assimilation to Enhance the Spatial Impact of Argo Floats in the Copernicus Mediterranean Biogeochemical Model." *Ocean Science*, 2024. DOI: 10.5194/os-20-689-2024
Scopus: Oceanography, quartile Q1

[J6] (a) A. Leporati, L. Manzoni, G. Mauri, G. Pietropolli, C. Zandron. "Inferring P Systems from Their Computing Steps: An Evolutionary Approach." *Swarm and Evolutionary Computation*, 2023. DOI: 10.1016/j.swevo.2022.101223
Scopus: General Computer Science, quartile Q1

[J5] G. Pietropolli, L. Manzoni, G. Cossarini. "Multivariate Relationship in Big Data Collection of Ocean Observing System." *Applied Sciences*, 2023. DOI: 10.3390/app13095634
Scopus: Computer Science Applications, quartile Q2

[J4] G. Nadizar*, G. Pietropolli*. "A Grammatical Evolution Approach to the Automatic Inference of P Systems." *Journal of Membrane Computing*, 2023. DOI: 10.1007/s41965-023-00125-w
Scopus: Applied Mathematics, quartile Q1

[J3] G. Pietropolli, G. Menara, M. Castelli. "A Genetic Programming Based Heuristic to Simplify Rugged Landscapes Exploration." *Emerging Science Journal*, 2023. DOI: 10.28991/ESJ-2023-07-04-01
Scopus: Multidisciplinary, quartile Q1

[J2] G. Pietropolli, L. Manzoni, A. Paoletti, M. Castelli. "On the Hybridization of Geometric Semantic GP with Gradient-Based Optimizers." *Genetic Programming and Evolvable Machines*, 2023. DOI: 10.1007/s10710-023-09463-1
Scopus: Computer Science Applications, quartile Q2

[J1] (a) M. Castelli, L. Manzoni, L. Mariot, G. Menara, G. Pietropolli. "The Effect of Multi-Generational Selection in Geometric Semantic Genetic Programming." *Applied Sciences*, 2022. DOI: 10.3390/app12104836
Scopus: Computer Science Applications, quartile Q2

Book Chapters

[B2] M. Castelli, G. Pietropolli, L. Manzoni. "Supervised Learning: Classification" *Encyclopedia of Bioinformatics and Computational Biology (Second Edition)*, Elsevier, 2025, pp.417-425. DOI: 10.1016/B978-0-323-95502-7.00116-0

[B1] L. Vanneschi, D. Farinati, D. Rasteiro, L. Rosenfeld, G. Pietropolli, S. Silva. "Exploring non-bloating geometric semantic genetic programming." *Genetic Programming Theory and Practice XXI*, Springer, 2024, pp.237-258. DOI: 10.1007/978-981-96-0077-9_12

Conference Proceedings (peer reviewed)

The list of publications is provided together with the current rank of the journal in Icore. An asterisk (*) indicates multiple first authors with equal contribution.

[C11] L. Rosenfeld, D. Farinati, D. Rasteiro, G. Pietropolli, K. B. Rebuli, S. Silva, L. Vanneschi. "Slim_gsgp: A Python Library for Non-Bloating GSGP." *Proceedings of the Genetic and Evolutionary Computation Conference*, July 2025. DOI: 10.1145/3712256.3726398
Icore: rank A

[C10] R. Ascone, L. Mariot, L. Manzoni, G. Pietropolli. "Evolving Cryptographic Boolean Functions with Reaction Systems." *Proceedings of the Genetic and Evolutionary Computation Conference*, July 2025. DOI: 10.1145/3712255.3726685
Icore: rank A

[C9] G. Pietropolli*, D. Farinati*, L. Manzoni, M. Castelli, S. Silva, L. Vanneschi. "Introducing Crossover in SLIM-GSGP." *European Conference on Genetic Programming (Part of EvoStar)*, April 2025. DOI: 10.1007/978-3-031-89991-1_7
Icore: rank B

[C8] L. Rovito, L. Bonin, D. Farinati, L. Vanneschi, L. Manzoni, A. De Lorenzo, G. Pietropolli. "Exploring the Integration of Cellular Structures in Genetic Programming-Based Methods." *European Conference on Genetic Programming (Part of EvoStar)*, April 2025. DOI: 10.1007/978-3-031-89991-1_8
Icore: rank B

[C7] D. Farinati*, G. Pietropolli*, L. Vanneschi. "Exploring the Impact of Data Scale on Mutation Step Size in SLIM-GSGP." *European Conference on Genetic Programming (Part of EvoStar)*, April 2025. DOI: 10.1007/978-3-031-89991-1_3
Icore: rank B

[C6] S. Jorgensen*, G. Nadizar*, G. Pietropolli*, L. Manzoni, E. Medvet, U. M. O'Reilly, E. Hemberg. "Large Language Model-based Test Case Generation for GP Agents." *Proceedings of the Genetic and Evolutionary Computation Conference*, July 2024. DOI: 10.1145/3638529.3654056
Icore: rank A

[C5] G. Pietropolli, S. Nichele, E. Medvet. "The Role of the Substrate in CA-based Evolutionary Algorithms." *Proceedings of the Genetic and Evolutionary Computation Conference*, July 2024. DOI: 10.1145/3638529.3654112
Icore: rank A

[C4] J. Ferreira, M. Castelli, L. Manzoni, G. Pietropolli. "A Self-Adaptive Approach to Exploit Topological Properties of Different GAs' Crossover Operators." *European Conference on Genetic Programming (Part of EvoStar)*, March 2023. DOI: 10.1007/978-3-031-29573-7_1
Icore: rank B

[C3] G. Pietropolli, F. J. Camerota Verdù, L. Manzoni, M. Castelli. "Parametrizing GP Trees for Better Symbolic Regression Performance through Gradient Descent." *Proceedings of the Genetic and Evolutionary Computation Conference*, July 2023. DOI: 10.1145/3583133.3590574
Icore: rank A

[C2] G. Pietropolli, L. Manzoni, A. Paoletti, M. Castelli. "Combining Geometric Semantic GP with Gradient-Descent Optimization." *European Conference on Genetic Programming (Part of EvoStar)*, April 2022. DOI: 10.1007/978-3-031-02056-8_2
Icore: rank B

[C1] G. Pietropolli, G. Cossarini, L. Manzoni. "GANs for Integration of Deterministic Model and Observations in Marine Ecosystem." *EPIA Conference on Artificial Intelligence*, September 2022. DOI: 10.1007/978-3-031-16474-3_37

Workshop Papers (peer reviewed)

[W2] F.J.J.B. Santos, A. De Lorenzo, L. Manzoni, G. Pietropolli. "Learning the Particle Swarm Optimization Velocity Update via Genetic Programming." *Proceedings of the Genetic and Evolutionary Computation Conference*, July 2025. DOI: 10.1145/3712255.3734324

[W1] T. Tonelli, G. Pietropolli, G. Sbaiz, L. Manzoni. "Genetic Programming for the Reconstruction of Delay Differential Equations in Economics". *Proceedings of the Genetic and Evolutionary Computation Conference*, July 2024. DOI: 10.1145/3638530.3664164

Abstract and Techincal Reports

[A9] G. Pietropolli, C. Amadio, G. Cossarini, L. Manzoni. "GLOBIO: Bridging Global and Local Scales for Biogeochemical Profiles Prediction." *European Geoscience Union General Assembly*, May 2025.

[A8] T. Tonelli, G. Pietropolli, G. Cossarini, L. Manzoni. "Convolutional neural networks for chlorophyll prediction in the Mediterranean Sea." *European Geoscience Union General Assembly*, May 2025.

[A7] L. Trinchera, G. Pietropolli, M. Castelli, F. Schuberth. "Automated Specification Search for Composite-Based SEM." *Meeting of the Working Group Structural Equation Modeling (SEM)*, March 2025.

[A6] V. Blasone, U. Di Laudo, G. Pietropolli, L. Bortolussi, S. Ceramicola, G. Cossarini, L. Manzoni. "Machine Learning methods for the Atmosphere, the Ocean, and the Seabed" *secondo Convegno Nazionale CINI sull'Intelligenza Artificiale* - Local Proceedings, June 2023.

[A5] G. Pietropolli, L. Manzoni, G. Cossarini. "A convolutional deep learning approach for the reconstruction of nutrient and carbonate system variable profiles." *54th International Liege Colloquium On Ocean Dynamics*, May 2023.

[A4] C. Amadio, A. Teruzzi, G. Pietropolli, L. Manzoni, G. Coidessa, G. Cossarini. "Combining Neural Networks and Data Assimilation to enhance the spatial impact of Argo floats in the Copernicus Mediterranean biogeochemical model." *54th International Liege Colloquium On Ocean Dynamics*, May 2023.

[A3] M.E. Fabri, G. Pietropolli, L. Manzoni. "P Systems inference via Grammatical Evolution." *23rd Conference on Membrane Computing* – Local Proceedings, Sep 2022.

[A2] G. Nadizar, G. Pietropolli. "Construction of semi-uniform membrane structures in a uniform way." *23rd Conference on Membrane Computing* – Local Proceedings, Sep 2022.

[A1] L. Manzoni, G. Pietropolli, G. Cossarini. "Data Collection and AI-Assisted Modeling of Oceans" *secondo Convegno Nazionale CINI sull'Intelligenza Artificiale* - Local Proceedings, June 2022.

Privacy

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